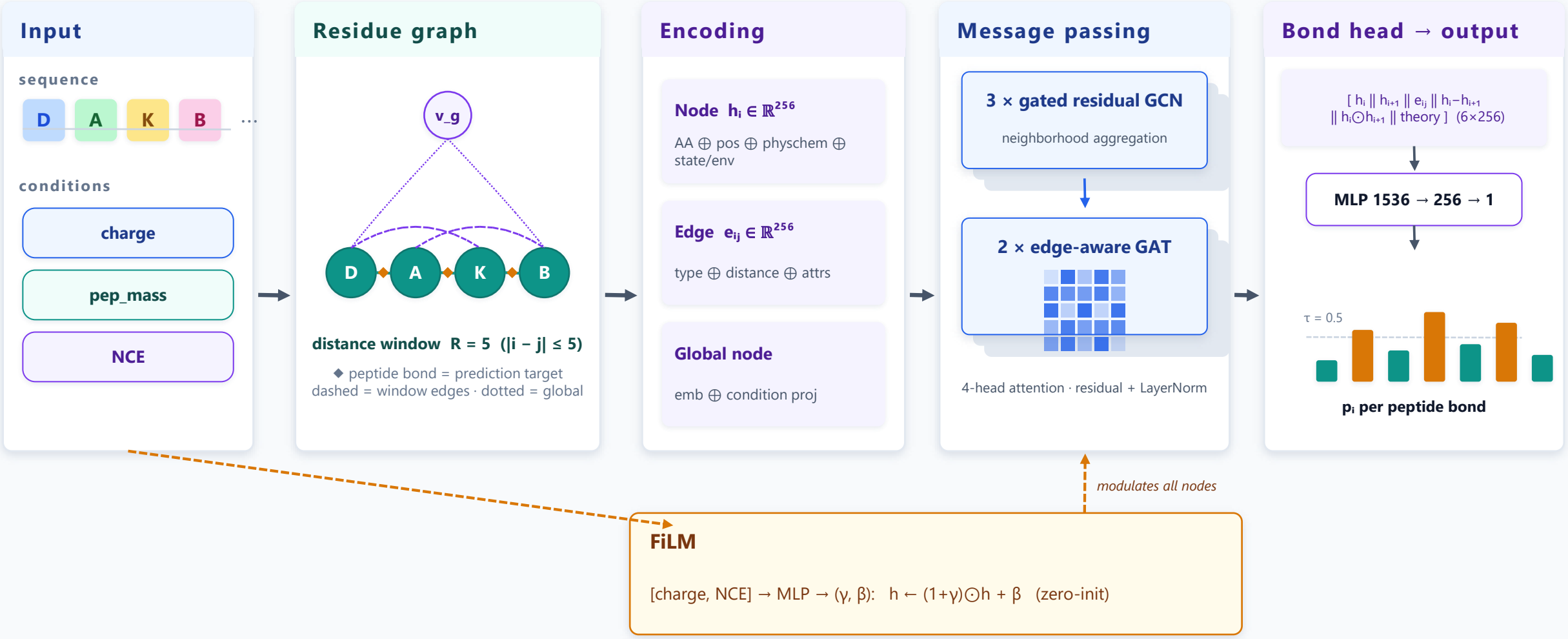

Input
sequence
D
A
K
B
...
conditions
charge
pep_mass
NCE
Residue graph
v_g
D
A
K
B
distance window R = 5 (|i − j| ≤ 5)
◆ peptide bond = prediction target
dashed = window edges · dotted = global
Encoding
Node $h_i \in \mathbb{R}^{256}$
AA ⊕ pos ⊕ physchem ⊕ state/env
Edge $e_{ij} \in \mathbb{R}^{256}$
type ⊕ distance ⊕ attrs
Global node
emb ⊕ condition proj
Message passing
3 × gated residual GCN
neighborhood aggregation
2 × edge-aware GAT
4-head attention · residual + LayerNorm
Bond head → output
[h_i ‖ h_{i+1} ‖ e_{ij} ‖ $h_i - h_{i+1}$ ‖ $h_i \odot h_{i+1}$ ‖ theory] (6×256)
MLP 1536 → 256 → 1
τ = 0.5
p_i per peptide bond
modulates all nodes
FiLM
[charge, NCE] → MLP → (γ, β): h ← (1+γ)⊙h + β (zero-init)